

Circuit Documentation

Summary

This circuit integrates multiple components to create a system capable of controlling motors, processing data from encoders, and interfacing with a Lidar sensor. The main components include an Arduino Uno R4 WiFi, a NodeMCU, a DRV8833 Motor Driver, two N20 Encoders, an RP Lidar A1M8, a DC-DC Buck Converter, and a Li-Ion Battery. The system is designed to manage motor control and data acquisition, with power supplied by a Li-Ion battery and regulated by a DC-DC converter.

Component List

1. **Arduino Uno R4 WiFi**
 - **Description:** A microcontroller board based on the ATmega328P, featuring WiFi connectivity.
 - **Pins:** 0/RX, 1/TX, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, GND, SCL, AREF, SDA, GPIO 42, GPIO 43, GPIO 44, GPIO 0, GPIO 41, VRTC, OFF, BOOT, IOREF, RESET, 3.3V, 5V, VIN, A0, A1, A2, A3, A4, A5, 3V3, IIC0_SDL, IIC0_SCL
2. **NodeMCU**
 - **Description:** A low-cost open-source IoT platform based on the ESP8266 WiFi module.
 - **Pins:** A0, RSV, SD3, SD2, SD1, CMD, SD0, CLK, GND, 3V3, EN, RST, Vin, D0, D1, D2, D3, D4, D5, D6, D7, D8, RX, TX
3. **DRV8833 Motor Driver**
 - **Description:** A dual H-bridge motor driver capable of driving two DC motors.
 - **Pins:** IN4, IN3, GND, VCC, IN2, IN1, ULT, OUT4, OUT3, EEP, OUT2, OUT1, solderpad, solderpad2
4. **DC-DC Buck LM2596 3A LED**
 - **Description:** A step-down voltage regulator module with an adjustable output.
 - **Pins:** VOUT+, VOUT-, VIN+, VIN-
5. **ENCODER N20 (x2)**
 - **Description:** A small DC motor with an integrated encoder for precise speed and position control.
 - **Pins:** M2, VCC, C2, C1, GND, M1
6. **RP Lidar A1M8 (old)**
 - **Description:** A 360-degree 2D laser scanner (Lidar) for mapping and navigation.
 - **Pins:** GND, RX, TX, V5.0, MOTOCTL, VMOTO
7. **Li-Ion Battery 7.4V 5200mAh**
 - **Description:** A rechargeable lithium-ion battery providing power to the circuit.
 - **Pins:** -VE, +VE

Wiring Details

Arduino Uno R4 WiFi

- **Pin 2:** Connected to C1 of ENCODER N20 (Left Motor)
- **Pin 3:** Connected to C1 of ENCODER N20 (Right Motor)
- **Pin 4:** Connected to C2 of ENCODER N20 (Left Motor)
- **Pin 5:** Connected to C2 of ENCODER N20 (Right Motor)
- **Pin 6:** Connected to IN1 of DRV8833 Motor Driver
- **Pin 8:** Connected to EEP of DRV8833 Motor Driver
- **Pin 9:** Connected to IN4 of DRV8833 Motor Driver
- **Pin 10:** Connected to IN3 of DRV8833 Motor Driver
- **Pin 11:** Connected to IN2 of DRV8833 Motor Driver
- **Pin 5V:** Connected to V5.0, MOTOCTL, VMOTO of RP Lidar A1M8, VCC of both ENCODER N20s, Vin of NodeMCU, and VOUT+ of DC-DC Buck LM2596 3A LED

- **Pin GND:** Connected to GND of RP Lidar A1M8, both ENCODER N20s, NodeMCU, and VOUT- of DC-DC Buck LM2596 3A LED

NodeMCU

- **Pin RX:** Connected to TX of RP Lidar A1M8
- **Pin TX:** Connected to RX of RP Lidar A1M8
- **Pin Vin:** Connected to V5.0, MOTOCTL, VMOTO of RP Lidar A1M8, VCC of both ENCODER N20s, 5V of Arduino Uno R4 WiFi, and VOUT+ of DC-DC Buck LM2596 3A LED
- **Pin GND:** Connected to GND of RP Lidar A1M8, both ENCODER N20s, Arduino Uno R4 WiFi, and VOUT- of DC-DC Buck LM2596 3A LED

DRV8833 Motor Driver

- **Pin IN1:** Connected to Pin 6 of Arduino Uno R4 WiFi
- **Pin IN2:** Connected to Pin 11 of Arduino Uno R4 WiFi
- **Pin IN3:** Connected to Pin 10 of Arduino Uno R4 WiFi
- **Pin IN4:** Connected to Pin 9 of Arduino Uno R4 WiFi
- **Pin OUT1:** Connected to M1 of ENCODER N20 (Left Motor)
- **Pin OUT2:** Connected to M2 of ENCODER N20 (Left Motor)
- **Pin OUT3:** Connected to M1 of ENCODER N20 (Right Motor)
- **Pin OUT4:** Connected to M2 of ENCODER N20 (Right Motor)
- **Pin EEP:** Connected to Pin 8 of Arduino Uno R4 WiFi
- **Pin VCC:** Connected to VIN+ of DC-DC Buck LM2596 3A LED and +VE of Li-Ion Battery
- **Pin GND:** Connected to VIN- of DC-DC Buck LM2596 3A LED and -VE of Li-Ion Battery

DC-DC Buck LM2596 3A LED

- **Pin VIN+:** Connected to +VE of Li-Ion Battery and VCC of DRV8833 Motor Driver
- **Pin VIN-:** Connected to -VE of Li-Ion Battery and GND of DRV8833 Motor Driver
- **Pin VOUT+:** Connected to V5.0, MOTOCTL, VMOTO of RP Lidar A1M8, VCC of both ENCODER N20s, 5V of Arduino Uno R4 WiFi, and Vin of NodeMCU
- **Pin VOUT-:** Connected to GND of RP Lidar A1M8, both ENCODER N20s, Arduino Uno R4 WiFi, and NodeMCU

ENCODER N20

- **M1:** Connected to OUT1 of DRV8833 Motor Driver
- **M2:** Connected to OUT2 of DRV8833 Motor Driver
- **C1:** Connected to Pin 2 of Arduino Uno R4 WiFi
- **C2:** Connected to Pin 4 of Arduino Uno R4 WiFi
- **VCC:** Connected to V5.0, MOTOCTL, VMOTO of RP Lidar A1M8, 5V of Arduino Uno R4 WiFi, Vin of NodeMCU, and VOUT+ of DC-DC Buck LM2596 3A LED
- **GND:** Connected to GND of RP Lidar A1M8, Arduino Uno R4 WiFi, NodeMCU, and VOUT- of DC-DC Buck LM2596 3A LED

ENCODER N20 (1)

- **M1:** Connected to OUT3 of DRV8833 Motor Driver
- **M2:** Connected to OUT4 of DRV8833 Motor Driver
- **C1:** Connected to Pin 3 of Arduino Uno R4 WiFi
- **C2:** Connected to Pin 5 of Arduino Uno R4 WiFi
- **VCC:** Connected to V5.0, MOTOCTL, VMOTO of RP Lidar A1M8, 5V of Arduino Uno R4 WiFi, Vin of NodeMCU, and VOUT+ of DC-DC Buck LM2596 3A LED
- **GND:** Connected to GND of RP Lidar A1M8, Arduino Uno R4 WiFi, NodeMCU, and VOUT- of DC-DC Buck LM2596 3A LED

RP Lidar A1M8 (old)

- **Pin TX:** Connected to RX of NodeMCU
- **Pin RX:** Connected to TX of NodeMCU
- **Pin V5.0:** Connected to MOTOCTL, VMOTO, VCC of both ENCODER N20s, 5V of Arduino Uno R4 WiFi, Vin of NodeMCU, and VOUT+ of DC-DC Buck LM2596 3A LED
- **Pin GND:** Connected to GND of both ENCODER N20s, Arduino Uno R4 WiFi, NodeMCU, and VOUT- of DC-DC Buck LM2596 3A LED

Li-Ion Battery 7.4V 5200mAh

- **Pin +VE:** Connected to VIN+ of DC-DC Buck LM2596 3A LED and VCC of DRV8833 Motor Driver
- **Pin -VE:** Connected to VIN- of DC-DC Buck LM2596 3A LED and GND of DRV8833 Motor Driver

